

1.3 JD+ and R

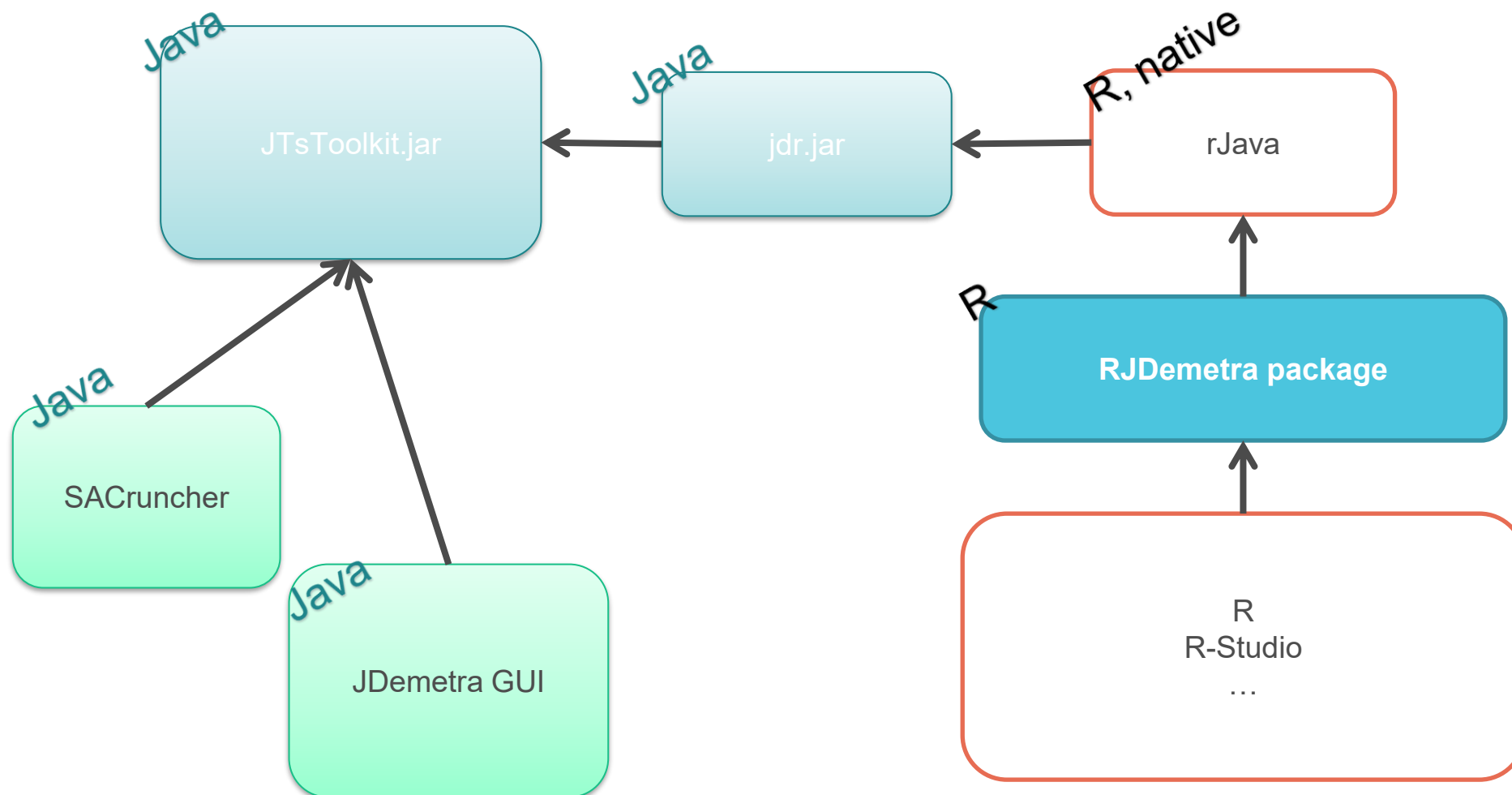
ESTP Training
05-07 October 2026

eurostat 

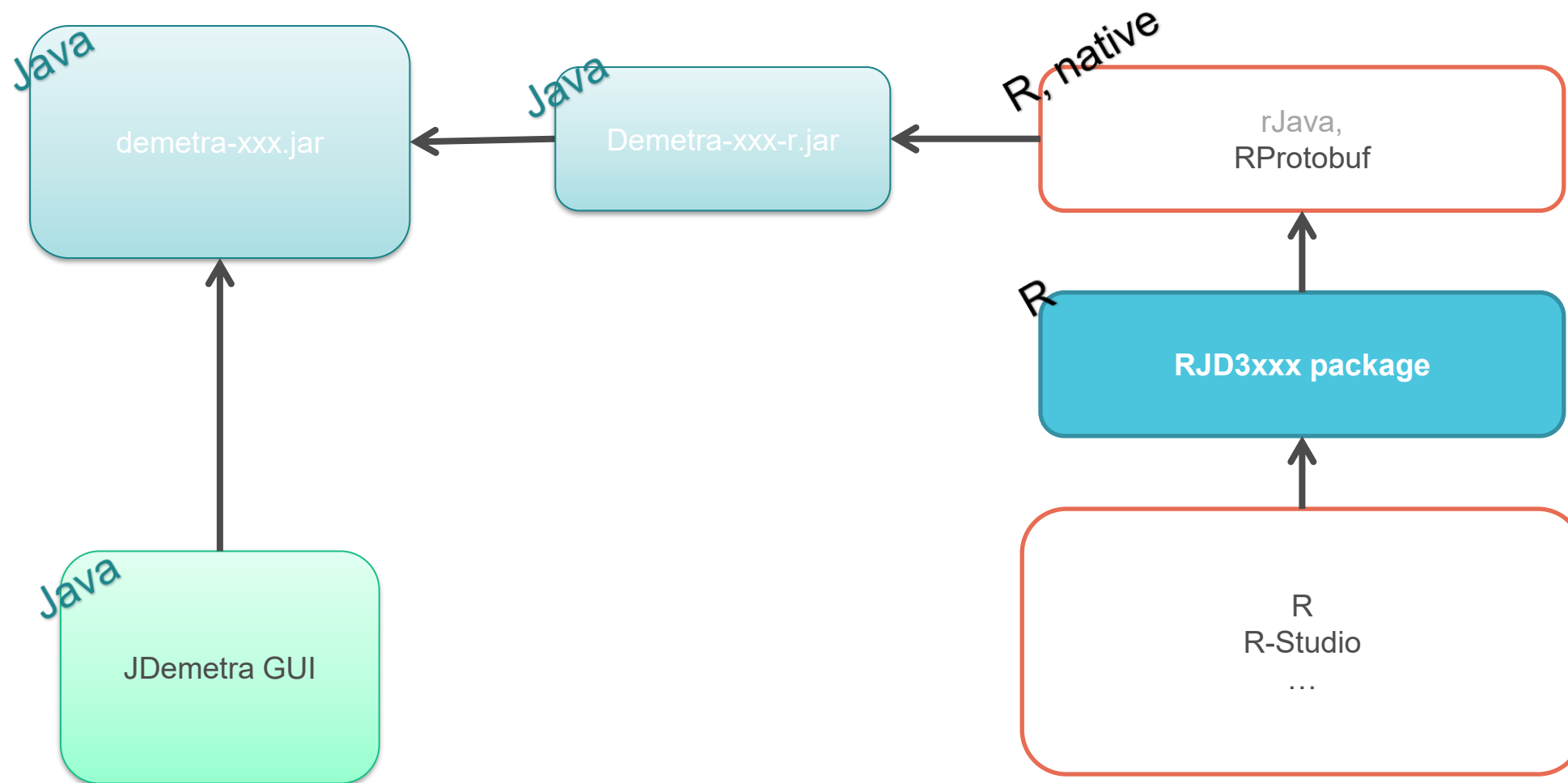
1. Main requirements

- RJDemetra
 - **Java runtime (≥ 11)**
 - R ($\geq 3.1.1$)
 - rJava ($\geq 0.9-8$)
- rjdverse
 - **Java runtime (≥ 17.0)**
 - R ($\geq 3.6.0$)
 - rJava ($\geq 1.0-6$),
 - RProtoBuf ($\geq 0.4.17$)

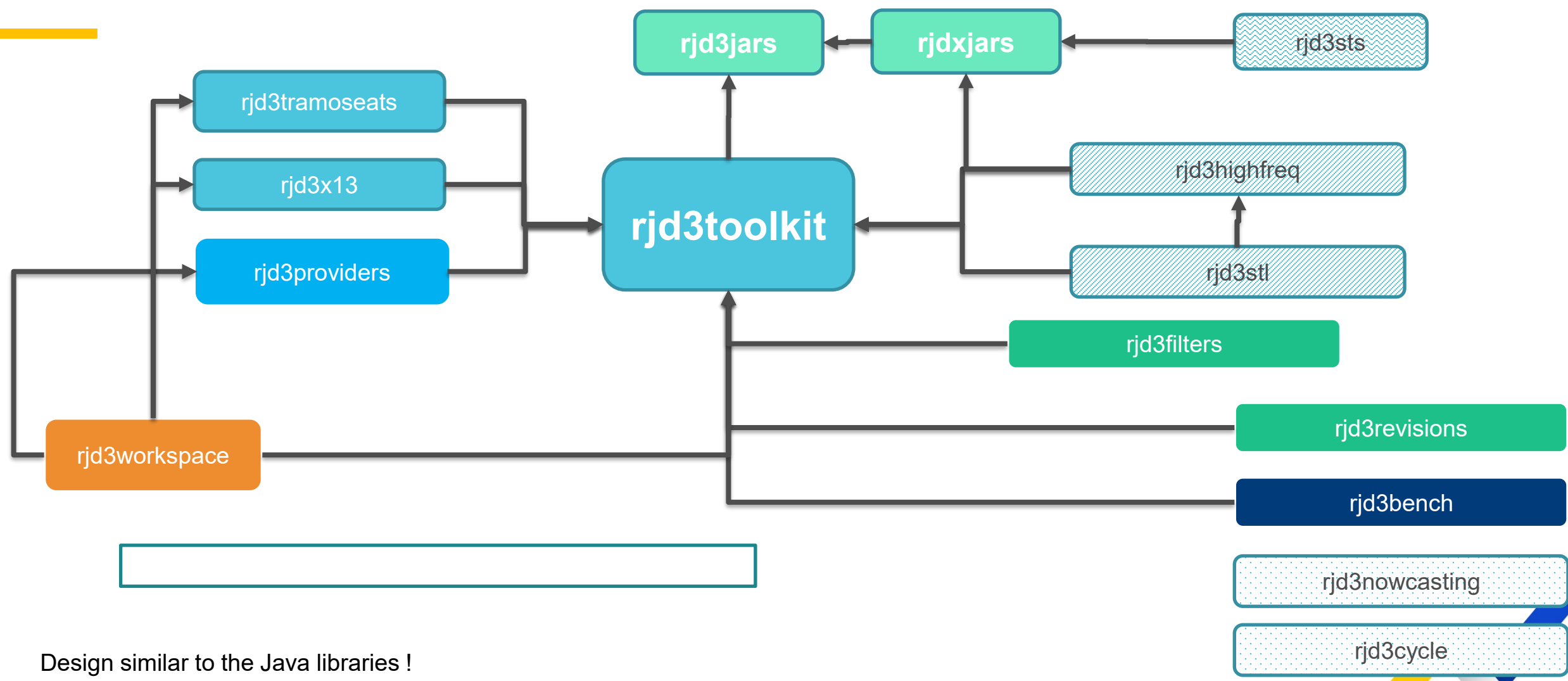
2. Technical design of Rjdemetra (v2)



3. Technical design of rjdverse (v3)

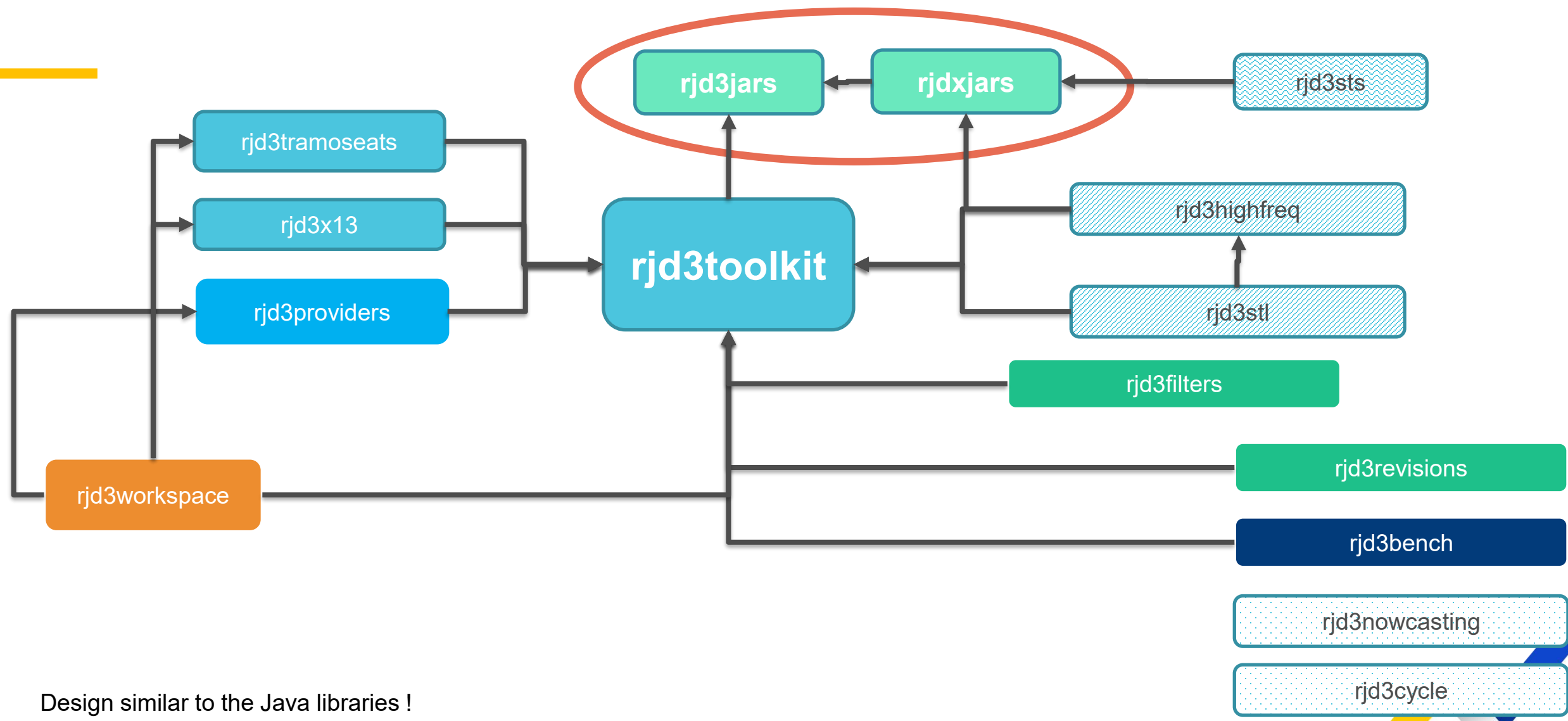


4. rjdverse: Overview (<https://github.com/rjdverse>)



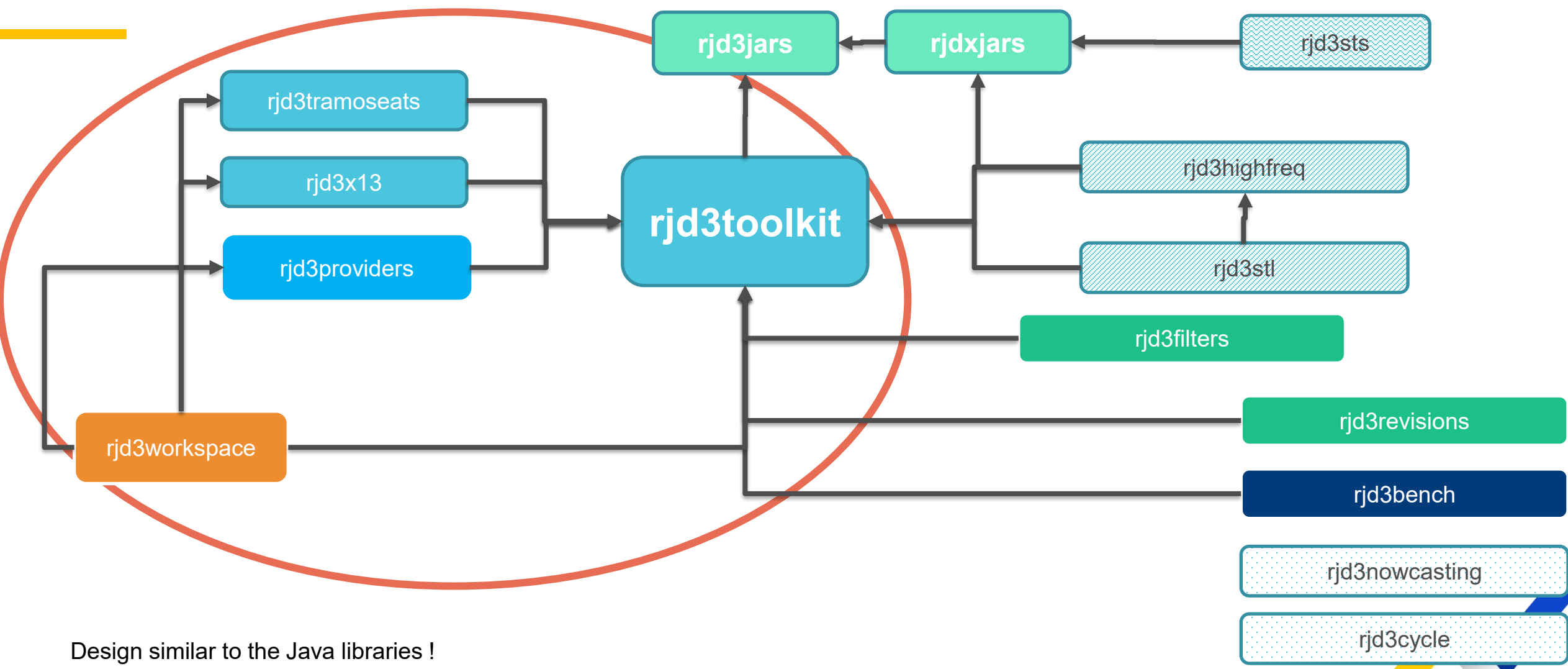
Design similar to the Java libraries !

4. rjdverse: Overview (Java libraries)

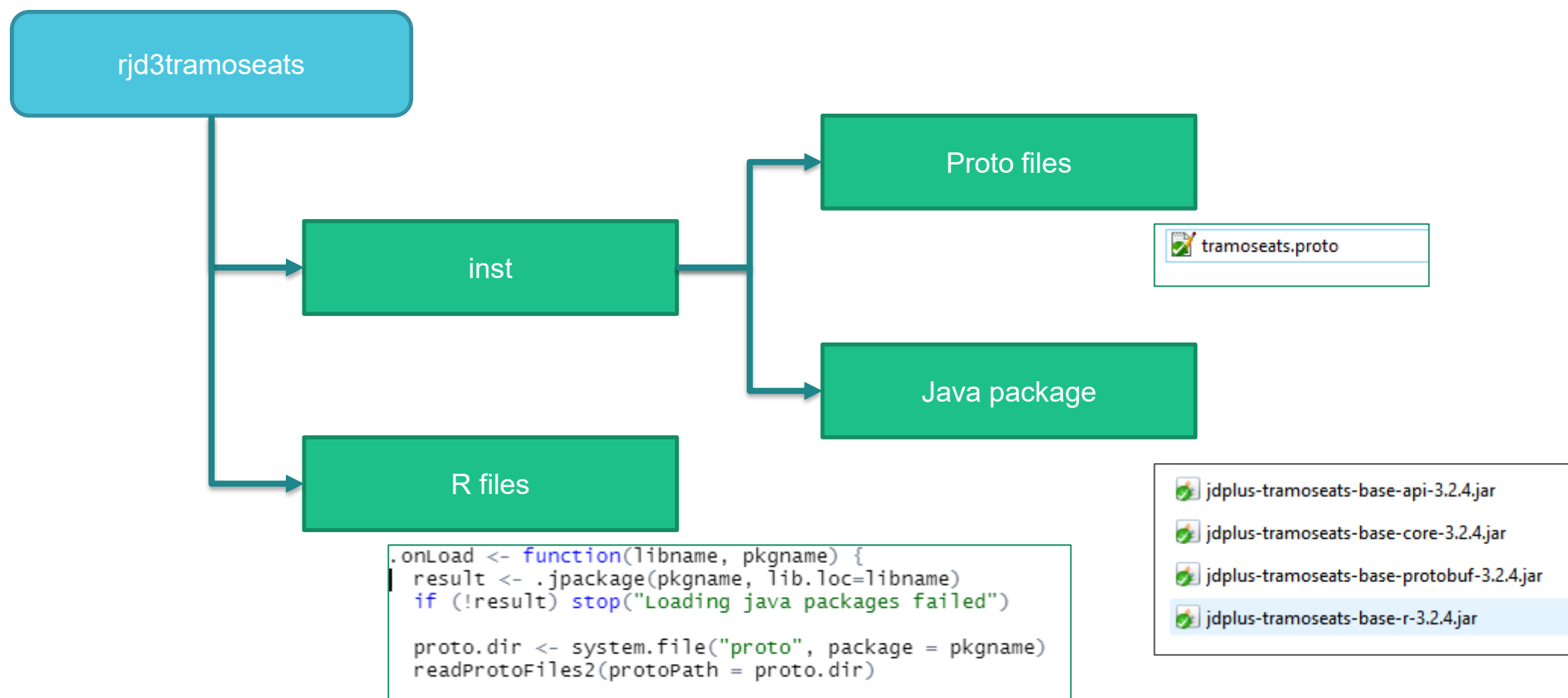


Design similar to the Java libraries !

4. rjdverse: Overview (main packages for SA)



Design similar to the Java libraries !



5. Objectives of R packages

- High-level functions with most common results
- Many low-level functions
 - Advanced users
 - Research
 - Training
 - Additional tools

6. Installing the packages

- If need be, referencing the correct java runtime (≥ 21.0)

```
R 4.4.0 · C:/Eurostat/ESTP/ESTP2024/R/estp_2024/
> usethis::edit_r_environ()
```

- Set your JAVA_HOME variable (tip: use the jre provided with JD+)

```
filters.Rmd × .Renviron* ×
1 JAVA_HOME='C:/Software/JD+/nbdemetra/jdk-21.0.3+9-jre'
```

```
Console Terminal × Background Jobs ×
R 4.4.0 · C:/Eurostat/ESTP/ESTP2024/R/estp_2024/
> Sys.getenv('JAVA_HOME')
[1] "C:/Software/JD+/nbdemetra/jdk-21.0.3+9-jre"
> |
```

- Install the various packages (internet access needed)
 - Install from CRAN for the main packages
 - `# install.packages("remotes")`
 - `remotes::install_github("rjdverse/rjd3workspace@*release")`

6. Installing the packages (cont)

- *remotes::install_github("rjdverse/rjd3filters@*release")*
- *remotes::install_github("rjdverse/rjd3sts@*release")*
- *remotes::install_github("rjdverse/rjd3highfreq@*release")*
- *remotes::install_github("rjdverse/rjd3x11plus@*release")*

7. Examples

- Reading Excel files (JD+-like) and detecting errors

```
rjd3providers::set_spreadsheet_paths('./Data')
print(rjd3providers::spreadsheet_content("belgium.xlsx"))

indprod<-rjd3providers::spreadsheet_data('belgium.xlsx', 1)
plot(indprod$series$`Manufacture of textiles`$data, col='blue')

err<-lapply(indprod$series, function(z)rjd3tramoseats::terror(z$data, 'tr1', nback=6))
```

- Refreshing a workspace

```
jws<-rjdemetra3::jws_load(system.file('workspaces', 'test.xml', package='rjdemetra3'))
ws<-rjdemetra3::read_workspace(jws)
jws2<-rjdemetra3::jws_make_copy(jws)
rjd3providers::set_spreadsheet_paths("c:/localdata/data/excel/new")
rjdemetra3::jws_refresh(jws2, 'Complete')
ws2<-rjdemetra3::read_workspace(jws2)

sa1<-ws$processing$`SAProcessing-1`$`Exports
France`
sa2<-ws2$processing$`SAProcessing-1`$`Exports
France`
ts.plot(ts.union(sa1$results$final$sa$data, sa2$results$final$sa$data), col=c('red', 'blue'))
print(window(sa2$results$final$series$data-sa1$results$final$series$data, start=2018))
```

8. Final remarks

- Many features provided in the Java libraries can be called from R
- Most tasks can be automated
- Many additional tools could be developed in R

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